

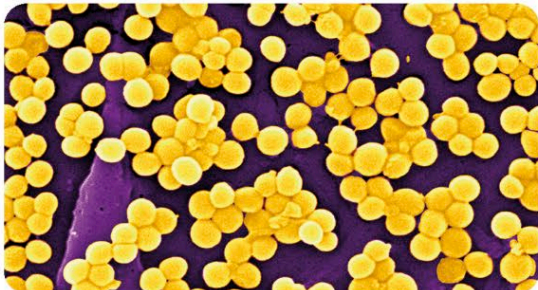
DEFEATING DISEASES

ANTIBIOTICS

Bacteria are tiny organisms that can cause diseases such as pneumonia. However, some bacteria and fungi (another type of organism) naturally produce substances to fight off and kill competing bacteria. Scientists use these to create antibiotics, which work as medicine to treat diseases that result from bacterial infections. More than half of the antibiotics in use today come from different species of the bacteria called *Streptomyces*, which can grow in large colonies.

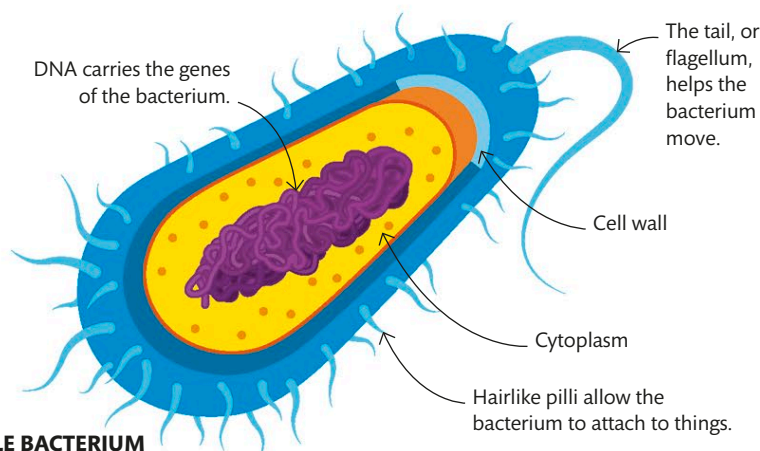
ANTIBIOTIC RESISTANCE

Bacteria that are in contact with antibiotics for too long can evolve the ability to resist them. Antibiotic-resistant bacteria, such as methicillin-resistant *Staphylococcus aureus* (MRSA) seen here, can cause dangerous infections that cannot be treated.



BACTERIA

Bacteria are single-celled organisms and are the most common living things. Most bacteria are helpful. Around 40 trillion of them live in your body, and the bacteria in your gut help you digest food. Some types are harmful and can cause infection. Antibiotics work in different ways to kill bacteria or prevent them from working.



A SINGLE BACTERIUM

These droplets are used to make the antibiotic.

